Maximum Likelihood Estimation

Yizhuo Hu

December 6th, 2024

1 Probability VS. Likelihood

The differences between the likelihood function and the probability density function are nuanced but important.

A probability density function $P(X|\theta)$: the probability of observing our data (X) given the underlying distribution parameters. It assumes that the parameters are **known**.

The likelihood function $L(\theta|X)$: the likelihood of parameter values (θ) occurring given the observed data. It assumes that the parameters are **unknown**.

Scenario Description: An Example with Temperature and Clothes

Parameter (θ): Represents the environmental condition, such as temperature (e.g., whether it is hot or cold).
Observation (X): Represents the observed outcome, such as the type of clothes people are wearing (e.g., summer clothes).
Mathematical Relationship:

$$P(X|\theta) \quad \leftrightarrow \quad L(\theta|X)$$

- Probability ($P(X|\theta)$): Given the parameter θ (e.g., high temperature), we predict the probability of observing a specific outcome (e.g., people wearing summer clothes).

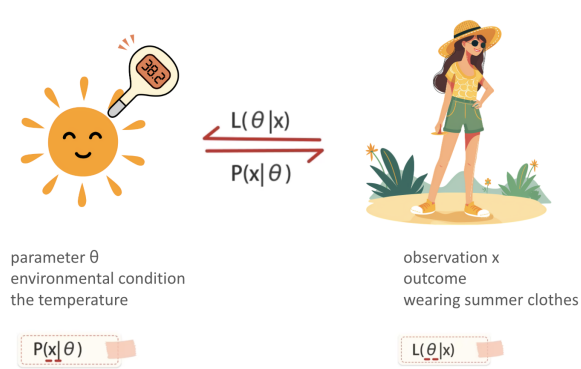

Figure 1

- Likelihood ($L(\theta|X)$): Given the observed outcome X (e.g., people wearing summer clothes), we infer the possible value of the parameter θ (e.g., high temperature).

2 Intuition

- The intuition of the maximum likelihood principle is as follows.
- From the (assumed) distribution of the data (e.g. wearing summer clothes), we determine the likelihood of observing the sample that we happen to observe as a function of the unknown parameters (e.g. high tempature) that characterize the distribution.
- Next, we choose as our maximum likelihood estimates those values for the unknown parameters that give us the highest likelihood.

2.1 a. Observed Data (X)

The data you observe is that people are wearing summer clothes. This is your real-world evidence.

2.2 b. Model Assumption ($P(X|\theta)$)

You assume that the probability of people wearing summer clothes depends on the temperature. For example:

- If the temperature is high (e.g., $35°C$), there's a high probability ($P(X|\theta)$) of seeing people in summer clothes.
- If the temperature is low (e.g., $10°C$), the probability of seeing summer clothes is low.

2.3 c. Likelihood ($L(\theta|X)$)

MLE asks: *Given the observed data (people wearing summer clothes), which temperature (θ) makes this data most probable?*

- For example, if $35°C$ makes it 95% likely for people to wear summer clothes and $10°C$ makes it only 5% likely, $35°C$ is the "most likely" temperature.

2.4 d. Maximizing the Likelihood

You calculate the likelihood for different possible temperatures (θ) and choose the value of θ that maximizes $L(\theta|X)$. This is the temperature that best explains the observed data.

2.5 Summary

MLE is like being a detective: you observe clues (summer clothes) and infer the most probable cause (high temperature). It formalizes this process by finding the parameter value (θ) that maximizes the likelihood of the observed data (X).

3 General Properties and Notation

The starting point of MLE is the assumption that the (conditional) distribution of an observed phenomenon (the endogenous variable (X) is known, except for a finite number of unknown parameters (θ). These parameters will be estimated by taking those values for them that give the observed values the highest probability, the highest likelihood. The maximum likelihood method thus provides a means of estimating a set of parameters characterizing a distribution, if we know, or assume we know, the form of this distribution.

The principle of maximum likelihood yields a choice of the estimator $\hat{\theta}$ as the value for the parameter that makes the observed data most probable.

- Suppose our interest is in the conditional distribution of y given $\mathbf{X}$.

  Let the density be given by $f(y_i|\mathbf{x}_i;\boldsymbol{\theta})$, where $\boldsymbol{\theta}$ is a K-dimensional vector of unknown parameters.

  With independent observations, the likelihood function for the entire sample is given by

$$L(\boldsymbol{\theta}|\mathbf{y},\mathbf{X}) = \prod_{i=1}^{N} L_i(\boldsymbol{\theta}|y_i,\mathbf{x}_i) = \prod_{i=1}^{N} f(y_i|\mathbf{x}_i;\boldsymbol{\theta}).$$

Log-Likelihood Function and Maximum Likelihood Estimator

- Typically, maximizing the *score function*, $\log L(\boldsymbol{\theta}|\mathbf{y},\mathbf{X})$, the logarithm of the likelihood, will be easier. The log-likelihood function is given by:

$$\log L(\boldsymbol{\theta}|\mathbf{y},\mathbf{X}) = \sum_{i=1}^{N} \log L_i(\boldsymbol{\theta}|y_i,\mathbf{x}_i) = \sum_{i=1}^{N} \log f(y_i|\mathbf{x}_i;\boldsymbol{\theta}).$$

- The ML estimator solves:

$$\max_{\boldsymbol{\theta}} \log L(\boldsymbol{\theta}) = \max_{\boldsymbol{\theta}} \sum_{i=1}^{N} \log L_i(\boldsymbol{\theta}).$$

First-Order Conditions for ML Estimator

- The first-order conditions imply:

$$\left.\frac{\partial \log L(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right|_{\hat{\boldsymbol{\theta}}} = \sum_{i=1}^{N} \left.\frac{\partial \log L_i(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right|_{\hat{\boldsymbol{\theta}}} = 0.$$

Score Vector and First-Order Conditions

- The vector of first derivatives of the log-likelihood function is also known as the **score vector**:

$$\mathbf{s}(\boldsymbol{\theta}) \equiv \frac{\partial \log L(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \sum_{i=1}^{N} \frac{\partial \log L_i(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \sum_{i=1}^{N} \mathbf{s}_i(\boldsymbol{\theta}).$$

- Thus, the first-order conditions:

$$\mathbf{s}(\hat{\boldsymbol{\theta}}) = \sum_{i=1}^{N} \mathbf{s}_i(\hat{\boldsymbol{\theta}}) = 0$$

  imply that the sum (or the sample average) of the score contributions evaluated at $\hat{\boldsymbol{\theta}}$ should be zero.

3.1 Properties of the ML Estimator

We will learn that especially for large samples, the maximum likelihood estimators have many desirable properties. Under weak regularity conditions, it can be shown that:

- The ML estimator is consistent;
- The ML estimator is asymptotically efficient;
- The ML estimator has an asymptotic normal distribution:

$$\sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \xrightarrow{d} \mathcal{N}(0, \mathbf{V}),$$

  where $\mathbf{V}$ is the asymptotic covariance matrix.

This class of estimators has an important property. If $\hat{\theta}(\mathbf{x})$ is a maximum likelihood estimate for θ, then $g(\hat{\theta}(\mathbf{x}))$ is a maximum likelihood estimate for $g(\theta)$. For example, if θ is a parameter for the variance and $\hat{\theta}$ is the maximum likelihood estimator, then $\sqrt{\hat{\theta}}$ is the maximum likelihood estimator for the standard deviation. This flexibility in estimation criterion seen here is not available in the case of unbiased estimators.

3.2 High Dimension Data

However, especially for high dimensional data, the likelihood can have many local maxima. Thus, finding the *global maximum* can be a major computational challenge. High-dimensional models in Maximum Likelihood Estimation (MLE) arise when the number of parameters (p) exceeds or is comparable to the number of observations (n), i.e., $p \gg n$. This situation leads to several challenges in model estimation and inference:

- **Overfitting:** When $p \gg n$, the model often fits the noise in the data, resulting in poor generalization performance.
- **Singularity:** In multivariate models, the covariance matrix becomes singular or non-invertible due to the insufficient number of observations.
- **Computational Complexity:** Solving the optimization problem in high dimensions becomes computationally expensive, especially when the likelihood function involves complex dependencies.
- **Unstable Estimates:** Parameter estimates become unstable due to high correlations or redundancies among features.

3.3 Reducing High Dimensionality in MLE: Recalling back the Seminar from Heqing

3.3.1 1. Ledoit-Wolf Shrinkage Estimation

Ledoit and Wolf (2003) propose a shrinkage approach to stabilize the covariance matrix estimation. The shrinkage estimator combines the sample covariance matrix S with a structured covariance matrix F:

$$\Sigma_{\text{Shrink}} = kF + (1-k)S,$$

where k is the shrinkage parameter that minimizes the estimation error. This approach is particularly useful in high-dimensional Gaussian models where the covariance matrix is required for the MLE computation.

3.3.2 2. Principal Orthogonal Complement Thresholding (POET)

Fan et al. (2013) introduced the POET method, which leverages factor models to reduce dimensionality. The sample covariance matrix S is decomposed using spectral decomposition:

$$S = \sum_{i=1}^{K} \lambda_i \xi_i \xi_i^\top + R_K,$$

where K is the number of principal components, λ_i are eigenvalues, ξ_i are eigenvectors, and R_K represents residuals. Thresholding is applied to R_K to remove noise, resulting in the regularized covariance matrix:

$$\hat{\Sigma}_K = \sum_{i=1}^{K} \lambda_i \xi_i \xi_i^\top + R_K^T.$$

3.3.3 3. Principal Component Analysis (PCA)

PCA is a general dimensionality reduction technique that transforms the original features into orthogonal principal components. By retaining only the top K components that explain the majority of the variance, the dimensionality is significantly reduced, allowing MLE to be computed in the reduced space.

3.3.4 4. Shrinkage Combined with Factor Models

The Ledoit-Wolf shrinkage and POET methods can be combined for better performance. For example, the structured covariance matrix F in the Ledoit-Wolf method can be derived from a factor model, capturing low-rank structures and stabilizing the estimation process.

3.3.5 5. Others

Apart from the methods mentioned in the seminar, several other approaches can be employed to address high dimensionality in Maximum Likelihood Estimation (MLE):

Regularization Methods Regularization introduces a penalty term to the likelihood function to control overfitting and reduce effective dimensionality.

- **Lasso (L1 Regularization):** Encourages sparsity in the parameter estimates by adding an $L1$-penalty to the likelihood:

  $$\ell_{\text{penalized}}(\theta) = \ell(\theta) - \lambda\|\theta\|_1.$$

  Many parameters are shrunk to exactly zero, effectively reducing dimensionality.

- **Ridge (L2 Regularization):** Shrinks parameter estimates by adding an $L2$-penalty:

  $$\ell_{\text{penalized}}(\theta) = \ell(\theta) - \lambda\|\theta\|_2^2.$$

  This method is particularly useful for handling multicollinearity but does not enforce sparsity.

- **Elastic Net:** Combines $L1$ and $L2$ penalties to balance sparsity and stability:

  $$\ell_{\text{penalized}}(\theta) = \ell(\theta) - \lambda_1\|\theta\|_1 - \lambda_2\|\theta\|_2^2.$$

Feature Selection Selecting relevant features before applying MLE simplifies the model and reduces dimensionality.

- **Filter Methods:** Use statistical measures such as correlation or mutual information to rank features and retain the most relevant ones.
- **Wrapper Methods:** Iteratively evaluate subsets of features by fitting the model and selecting those that optimize an objective function, such as AIC or cross-validation performance.
- **Embedded Methods:** Combine feature selection with model training, such as Lasso or tree-based models.

Subspace Learning These methods reduce dimensionality by projecting the data into a lower-dimensional subspace:

- **Kernel PCA:** Extends PCA to non-linear data by using kernel functions to project the data into a lower-dimensional space.
- **Linear Discriminant Analysis (LDA):** Projects data into a lower-dimensional space while maximizing class separability in classification problems.

Bayesian Methods Bayesian approaches incorporate prior knowledge to regularize the estimation process:

- Use sparsity-inducing priors (e.g., Laplace prior) to enforce sparsity in high-dimensional models.
- Approximate posterior distributions with variational inference or sampling-based methods like Markov Chain Monte Carlo (MCMC).

4 Examples

4.1 Example 1. Probability Distribution Estimation - Binomial

The principle of maximum likelihood is most easily introduced in a discrete setting where y_i only has a finite number of outcomes. As an example, consider a large pool filled with red and yellow balls. We are interested in the fraction p of red balls in this pool. To obtain information on p, we take a random sample of N balls (and do not look at all the other balls). Let us denote $y_i = 1$ if ball i is red and $y_i = 0$ if it is not. Then it holds by assumption that $P(y_i = 1) = p$. Suppose our sample contains $N_1 = \sum_i y_i$ red and $N - N_1$ yellow balls. The probability of obtaining such a sample (in a given order) is given by

$$P(N_1 \text{ red balls, } N - N_1 \text{ yellow balls}) = p^{N_1}(1-p)^{N-N_1} \tag{1}$$

The expression in (1), interpreted as a function of the unknown parameter p, is referred to as the likelihood function. Maximum likelihood estimation for p implies that we choose a value for p such that (1) is maximal. This gives the maximum likelihood estimator $\hat{p}$. For computational purposes, it is often more convenient to maximize the (natural) logarithm of (1), which is a monotone transformation. This gives the **loglikelihood function**

$$\log L(p) = N_1 \log(p) + (N - N_1)\log(1-p) \tag{2}$$

For a sample size of 100 with 44 red balls ($N_1 = 44$), 1 displays the loglikelihood function for values of p between 0.1 and 0.9. Maximizing (2) gives the first-order condition

$$\frac{d \log L(p)}{dp} = \frac{N_1}{p} - \frac{N - N_1}{1-p} = 0, \tag{3}$$

which, solving for p, gives the maximum likelihood (ML) estimator

$$\hat{p} = \frac{N_1}{N}. \tag{4}$$

The ML estimator thus corresponds to the sample proportion of red balls, and probably also corresponds to your best guess for p based on the sample that was drawn. In principle, we also need to check the second-order condition to make sure that the solution we have corresponds to a maximum, although in this case it is obvious from (1). This gives

$$\frac{d^2 \log L(p)}{dp^2} = -\frac{N_1}{p^2} - \frac{N - N_1}{(1-p)^2} < 0, \tag{5}$$

showing, indeed, that we have found a maximum.

4.2 Example 2. Simple Linear Regression

we could characterize the distribution of some variable y_i (for given x_i) as normal with mean $\beta_1 + \beta_2 x_i$ and variance σ^2. This would represent the simple linear regression model with normal error terms.

a. The Linear Regression Model

Consider the simple linear regression model:

$$y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$$

Under the Gauss-Markov assumptions and the assumption that the error is normally distributed, the **likelihood contribution** of observation i is given by the normal distribution function:

$$f(y_i|x_i;\beta,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2}\frac{(y_i-\beta_1-\beta_2 x_i)^2}{\sigma^2}\right\}.$$

b. Likelihood Function for N Observations

For N observations, assuming i.i.d., the joint density of $y_1, y_2, \ldots, y_N$ conditional on $X = (x_1, \ldots, x_N)'$ is given by:

$$\prod_{i=1}^{N} f(y_i|x_i;\beta,\sigma^2) = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2}\frac{(y_i-\beta_1-\beta_2 x_i)^2}{\sigma^2}\right\}.$$

This can be expressed as:

$$\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N \prod_{i=1}^{N} \exp\left\{-\frac{1}{2}\frac{(y_i-\beta_1-\beta_2 x_i)^2}{\sigma^2}\right\}.$$

This is the **likelihood function**, which is a function of the unknown parameters β and σ^2 — $L(\beta,\sigma^2)$.

c. The Log-Likelihood Function

The log-likelihood function is therefore:

$$\log L(\beta,\sigma^2) = -\frac{N}{2}\log(2\pi\sigma^2) - \frac{1}{2}\sum_{i=1}^{N}\frac{(y_i-\beta_1-\beta_2 x_i)^2}{\sigma^2}.$$

d. Maximizing the Likelihood Function

Since the first term $-\frac{N}{2}\log(2\pi\sigma^2)$ does not depend on (β_1) and (β_2), maximizing this likelihood function with respect to β is equivalent to minimizing the sum of squared residuals (Equivalent to OLS now):

$$min \sum_{i=1}^{N}(y_i-\beta_1-\beta_2 x_i)^2.$$

We obtain :

$$\hat{\beta} = \left(\sum_{i=1}^{N} x_i x_i'\right)^{-1}\left(\sum_{i=1}^{N} x_i y_i\right).$$

Maximizing $\log L$ with respect to σ^2 and letting $e_i = y_i - \beta_1 - \beta_2 x_i$, we obtain:

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{N} e_i^2}{N}.$$

e. Properties of the MLE Estimators

Note that $\hat{\beta}$ is identical to the OLS estimator b. It is consistent and also unbiased. $\hat{\sigma}^2$ is a consistent estimator for σ^2, but it is biased. Compare this to:

$$s^2 = \frac{\sum_{i=1}^{N} e_i^2}{N-K}.$$

The MLE underestimates the true variance because it does not account for the reduction in degrees of freedom.

4.3 Example 3. Binary Response Model

a. Introduction to Binary Response Models

We want to explain a binary outcome (e.g., yes or no), indicated by $y = 1$ or $y = 0$. **Econ Examples**: Does childbearing affect female labor supply? **Finance Examples**: Is CEO turnover sensitive to firm performance? Can a bank board prevent misconduct? Models boil down to specifying the probability that $y = 1$ as a function of $\mathbf{X}$.

b. Linear Probability Model (LPM) The **Linear Probability Model** assumes:

$$P\{y_i = 1|\mathbf{x}_i\} = \mathbf{x}_i'\beta$$

but restricts $\hat{y}_i$ to 0 if it is less than 0 and to 1 if it is greater than 1. Usually estimated using OLS with $\hat{\Sigma}_{HC}$. The LPM is generally robust and easy to interpret, but it has limitations (e.g., predictions can fall outside $[0, 1]$). β_k represents the impact of a change in x_{ik} on the probability of $y_i = 1$, an estimate of the **average treatment effect**.

c. Binary Response Model Using Maximum Likelihood (ML)

We specify the probability that $y_i = 1$ using $F(\mathbf{x}_i'\beta)$, where $F(\cdot)$ is a cumulative distribution function (CDF):Domain of $F(\cdot)$: $(-\infty, +\infty)$. Range of $F(\cdot)$: $[0, 1]$. $F(\cdot)$ is monotonically increasing. The likelihood function is:

$$L(\beta) = \prod_{i=1}^{N} L_i(\beta), \quad \text{where} \quad L_i(\beta) = \begin{cases} F(\mathbf{x}_i'\beta), & \text{if } y_i = 1, \\ 1 - F(\mathbf{x}_i'\beta), & \text{if } y_i = 0. \end{cases}$$

The log-likelihood function is:

$$\log L(\beta) = \sum_{i=1}^{N} \log L_i(\beta).$$

d. Maximum Likelihood Estimation

The Maximum Likelihood Estimator (MLE) for β is:

$$\mathbf{b}_{ML} = \arg\max_{\beta} \log L(\beta).$$

e. Marginal Effects in Probit Models

Once we obtain $\mathbf{b}_{ML}$ for the binary response model, the marginal effect is given by the derivative of $F(\mathbf{x}_i'\beta)$ with respect to x_{ik}:

$$\frac{\partial F(\mathbf{x}_i'\mathbf{b})}{\partial x_{ik}} = \beta_k F'(\mathbf{x}_i'\mathbf{b}),$$

where $F'(\cdot)$ is the probability density function (PDF) corresponding to $F(\cdot)$. For Probit:

$$F'(\cdot) = \phi(\cdot), \quad \text{the standard normal PDF.}$$

Important: β_k is not the marginal effect in these models; it is scaled by $F'(\mathbf{x}_i'\mathbf{b})$, which depends on the values of x_i.

5 Summary of Applications

Finance: MLE can help estimate parameters of models like Black-Scholes-Merton, which describe the dynamics of financial derivatives. Specifically, MLE can be used to calibrate a stochastic volatility model which is used to price options. This model involves a continuous-time stochastic differential equation with unknown parameters. By applying MLE on observed historical stock prices, these parameters can be estimated efficiently.

Category	**Description and Examples**
Probability Distribution Estimation	MLE is used to estimate parameters of known distributions (e.g., mean μ, variance σ^2): • Normal distribution: Estimate μ, σ^2. • Poisson distribution: Estimate rate λ. • Multivariate distributions: Estimate covariance matrices.
Regression Models	MLE is applied in regression analysis to estimate model parameters: • Linear regression: Estimate coefficients and variance. • Logistic regression: Estimate probabilities in binary classification. • Generalized linear models (GLMs): Handle non-normal data (e.g., Poisson or binomial).

Category	Description and Examples
Time Series Analysis	MLE is widely used in dynamic models: • ARIMA models: Estimate auto-regressive and moving average parameters. • GARCH models: Estimate volatility in financial time series. • State-space models: Estimate parameters for filtering (e.g., Kalman filters).
Classification and Prediction	MLE estimates parameters for classification models: • Logistic regression for binary classification. • Multinomial logistic regression for multi-class problems. • Naive Bayes classifier: Estimate conditional probabilities.
Clustering and Mixture Models	MLE supports clustering via: • Gaussian Mixture Models (GMMs): Estimate mixture weights, means, and variances using Expectation-Maximization (EM). • Other mixture models: E.g., Poisson mixtures.
Survival and Reliability Analysis	MLE is critical for time-to-event data: • Survival analysis: Estimate hazard rates and survival functions. • Reliability analysis: Estimate failure time distributions.

Category	Description and Examples
Model Comparison and Hypothesis Testing	MLE forms the foundation for: • Likelihood Ratio Tests (LRTs): Compare nested models. • Model selection criteria: AIC (Akaike Information Criterion), BIC (Bayesian Information Criterion).
Neural Networks and Deep Learning	Many loss functions in neural networks are derived from MLE: • Cross-entropy loss: Log-likelihood for classification. • Mean squared error: Log-likelihood for Gaussian regression.
Bayesian Analysis	MLE is a special case of Maximum A Posteriori (MAP) estimation with non-informative priors: • Provides initial parameter estimates for Bayesian models.
Signal Processing and Engineering Applications	MLE is widely used for: • Signal parameter estimation: Frequency, amplitude. • Positioning systems: E.g., GPS location estimation. • Direction finding: Estimating target directions in radar systems.
Anomaly Detection and Diagnosis	MLE supports identifying outliers or system failures: • Outlier detection: Based on low likelihood values. • Fault diagnosis: Estimate probabilities of system failures.

Category	Description and Examples
Discrete Choice Models	MLE is used in optimization problems: • Multinomial logit models: Estimate consumer choices. • Transportation models: Estimate preferences in route selection.

6 Advantages VS. Disadvantages

6.1 Advantages

- Maximum likelihood provides **a consistent approach to parameter estimation problems**. This means that maximum likelihood estimates can be developed for a large variety of estimation situations.
- Maximum likelihood methods have desirable mathematical and optimality properties. Specifically:
  - **They become minimum variance unbiased estimators as the sample size increases.** By unbiased, we mean that if we take (a very large number of) random samples with replacement from a population, the average value of the parameter estimates will be theoretically exactly equal to the population value.
  - By minimum variance, we mean that the estimator has the smallest variance, and thus the narrowest confidence interval, of all estimators of that type.
  - **They have approximate normal distributions** and approximate sample variances that can be used to generate confidence bounds and hypothesis tests for the parameters.

6.2 Disadvantages

- The likelihood equations **need to be specifically worked out for a given distribution and estimation problem**. The mathematics is often non-trivial, particularly if confidence intervals for the parameters are desired.
- **The numerical estimation is usually non-trivial**. Except for a few cases where the maximum likelihood formulas are in fact simple, it is generally best to rely on high-quality statistical software to obtain maximum likelihood estimates. Fortunately, high-quality maximum likelihood software is becoming increasingly common.
- Maximum likelihood estimates can be **heavily biased for small samples.** The optimality properties may not apply for small samples.

- Maximum likelihood can be sensitive to the choice of starting values (act as the initial position from which an algorithm begins its search for the optimal solution).

7 Programming Tools

R Language: **mle** function in package "**stats4**"
Python: $https://opensourceecon.github.io/CompMethods/struct_est/MLE.html$